Answer Key — Competitive Exam Practice 8: CPU Scheduling

GATE (Computer Science) Pattern

Auqib Hamid Lone

Department of Computer Science & Engineering, GCET Ganderbal

October 29, 2026

Concepts That Click

Q1 (A). A non-preemptive scheduler decides only when the CPU is released voluntarily: when the running process blocks (I) or terminates (IV). Decisions at II and III would take the CPU from a running process, which is preemption.

Q2 (B). With all processes at time 0, $\sum_i \mathrm{WT}_i = \sum_k (n-k)\tau_{(k)}$; the decreasing weights are matched to increasing bursts only by shortest-first order, so SJF is optimal. (D) is false: total work is fixed, but the average waiting time depends on the order (e.g. 17 vs. 3 ms for bursts 24, 3, 3).

Q3 (B). This is the convoy effect of non-preemptive FCFS: everyone moves at the speed of the long CPU-bound burst. Starvation (A) is the failure of SJF/SRTF, not this pattern.

Q4 (C). Expanding the recursion gives weights $\alpha, \alpha(1-\alpha), \alpha(1-\alpha)^2, \ldots$. (A) and (B) have the extremes swapped: $\alpha = 0$ keeps $\hat{\tau}_0$, $\alpha = 1$ copies the last burst. (D) is false: prediction does not prevent long jobs being passed over.

Q5 4.5. Gantt: `P1` 0–5, `P2` 5–8, `P3` 8–14, `P4` 14–15. $\mathrm{WT} = 0, 5-1, 8-3, 14-5 = 0, 4, 5, 9$; average $18/4 = 4.5$.

Q6 10. `P1` runs 0–8 (alone at 0). At 8 the ready bursts are 5, 2, 3: `P3` 8–10, `P4` 10–13, `P2` 13–18. $\mathrm{TAT} = 8, 18-2, 10-3, 13-4 = 8, 16, 7, 9$; average $40/4 = 10$.

Q7 2.25. At 2, `P2` (3) preempts `P1` (8 left); at 3, `P3` (1) preempts `P2` (2 left) and runs 3–4; `P2` 4–6; at 6, `P4` (4) beats `P1` (8) and runs 6–10; `P1` 10–18. $\mathrm{CT} = 18, 6, 4, 10$; $\mathrm{TAT} = 18, 4, 1, 4$; $\mathrm{WT} = 8, 1, 0, 0$; average $9/4 = 2.25$.

Q8 5. The Gantt sequence is `P1`, `P2`, `P3`, `P2`, `P4`, `P1`: six bars, hence five switches between different processes.

Q9 8. $\hat{\tau}_1 = 0.5(8) + 0.5(12) = 10$; $\hat{\tau}_2 = 0.5(6) + 0.5(10) = 8$.

Q10 5.33. Smallest (SJF order 2, 4, 10): $\mathrm{WT} = 0, 2, 6$, average $8/3 \approx 2.67$. Largest (order 10, 4, 2): $\mathrm{WT} = 0, 10, 14$, average $24/3 = 8$. Difference $8 - 8/3 = 16/3 \approx 5.33$ (accept 5.30–5.36). All six orders were checked; 8 is the maximum.